Complex Systems

Exam Preparation Worksheet (2025/26)

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Questions marked ★ are adapted from the compiled worksheet of past exam questions by Prof. Ernesto Costa. Questions marked ★ are taken or adapted from past written exams of the Complex Systems course (2020–2024). More than one ★ means multiple appearances.

Contents

Key Formulas Reference

Discrete Dynamical Systems

$$x_{n+1} = f(x_n) \qquad \text{fixed point: } x^* = f(x^*) \qquad \text{stable iff } |f'(x^*)| < 1$$

Logistic map: $x_{n+1} = r\,x_n(1-x_n)$, $x_n \in [0,1]$, $r \in [0,4]$. Bifurcation cascade: $r < 1$ extinct; $1 < r < 3$ stable FP; $3 < r < 3.57$ period doubling; $r \gtrsim 3.57$ chaos.

Continuous Dynamical Systems

$$\dot{x} = f(x) \qquad \text{fixed point: } f(x^*) = 0 \qquad \text{stable iff } f'(x^*) < 0$$

Euler: $x_{t+\Delta t} \approx x_t + \Delta t\, f(x_t)$, error $\propto \Delta t$. RK4: error $\propto (\Delta t)^4$.

Lotka-Volterra (predator-prey):

$$\dot{x} = ax - bxy \qquad \dot{y} = -cy + dxy$$

Lorenz system (3D, chaotic for $\sigma = 10$, $b = 8/3$, $\rho = 28$):

$$\dot{x} = \sigma(y-x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - bz$$

Chaos & Lyapunov

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln|f'(x_i)|$$

$\lambda > 0$: chaotic; $\lambda < 0$: periodic/stable; $\lambda = 0$: bifurcation boundary. Feigenbaum constant: $\delta \approx 4.669$.

Fractal Dimension

$$D = \frac{\log N_C}{\log M_F}$$

N_C = number of self-copies, M_F = magnification factor. Cantor set: $D \approx 0.631$; Sierpiński triangle: $D \approx 1.585$; Koch curve: $D \approx 1.262$.

Network Science

$$\text{degree centrality}: C_D(v) = \frac{k_v}{n-1} \qquad \text{closeness}: C_C(v) = \frac{n-1}{\sum_u d(v,u)} \qquad \text{betweenness}: C_B(v) = \sum_{s\neq v\neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

$$\text{clustering coeff}: c_v = \frac{2\,|\{e_{jk} : j,k \in N(v)\}|}{k_v(k_v-1)} \qquad \text{modularity}: Q = \sum_c \left[\frac{L_c}{m} - \left(\frac{d_c}{2m}\right)^2\right]$$

SIS model (network): $\rho_i(t+1) = (1-\mu)\rho_i + \beta(1-\rho_i)\sum_j A_{ij}\rho_j$

SIR model (ODE, well-mixed): $\dot{S} = -\beta SI/N$, $\dot{I} = \beta SI/N - \gamma I$, $\dot{R} = \gamma I$; epidemic grows iff $R_0 = \beta/\gamma > 1$.

Cellular Automata

$$\text{No. of rules} = q^{q^{2r+1}}$$

Binary ($q = 2$), radius $r = 1$: $2^8 = 256$ rules.

1 Concepts & Models

*A **complex system** exhibits emergent behavior that cannot be deduced from the properties of its parts in isolation. Key hallmarks: nonlinearity, feedback loops, emergence, self-organisation, and adaptation. We model such systems in three main ways: **mathematical models** (difference/differential equations), **agent-based models** (ABMs), and **network models**. No single approach is universally best — the choice depends on the type of question, available data, and level of detail required.*

1.1 ★★ Define *complex system*. What properties distinguish it from a merely "complicated" system?

1.2 ★★ Define *emergence*. Give an example from the course where emergent behavior arises from simple local rules.

1.3 List the three main classes of dynamic models studied in the course and briefly state for each: (i) what the state of the system is, (ii) how time evolves, (iii) one prototypical example studied in class.

1.4 ★★ Consider modeling the spread of an infectious virus. What are the advantages and disadvantages of each of the three modeling approaches (mathematical, agent-based, network)? Can these characteristics be generalised to other modeling problems?

1.5 What is the difference between a *deterministic* and a *stochastic* model? Give an example of each from the course.

1.6 Classify each of the following as discrete or continuous, linear or nonlinear, autonomous or non-autonomous, and state its order (dimension):

a) Bank account: $x_{t+1} = (1 + k)\, x_t$
b) Logistic map: $x_{t+1} = r\, x_t(1 - x_t)$
c) Lotka-Volterra: $\dot{x} = ax - bxy,\ \dot{y} = -cy + dxy$
d) Lorenz system

1.7 The **Boids** model of bird flocking is an early agent-based model. State its three rules and explain how global flocking behavior emerges from them. What concept in complex systems does this illustrate?

2 Discrete Dynamical Systems

*A **discrete dynamical system** is defined by a map $x_{n+1} = f(x_n)$. Its long-term behavior is governed by **fixed points** (x^* with $f(x^*) = x^*$) and their stability (determined by $|f'(x^*)|$). The **cobweb diagram** is a graphical iteration tool. The **logistic map** $x_{n+1} = rx_n(1 - x_n)$ is the canonical example: as r increases it undergoes a period-doubling **bifurcation cascade** leading to chaos. Multi-variable discrete systems are written as vector maps and can model coupled populations (predator-prey, Romeo & Juliet, stages-of-life).*

2.1 ★★★★ Define *fixed point* of a discrete dynamical system. Explain the graphical interpretation of a fixed point in a cobweb diagram.

2.2 ★ Define *cobweb diagram*. Describe the procedure for constructing it and explain what the shape of the spiral/staircase tells you about stability.

2.3 ★★★ Define *basin of attraction*. Is every fixed point guaranteed to have one? Explain.

2.4 ★★ Consider the discrete system $x_{n+1} = x_n^3$.

a) Find all fixed points.
b) Determine the stability of each fixed point.
c) Describe the long-term behavior for $x_0 \in (-1, 0) \cup (0, 1)$, for $|x_0| > 1$, and for $x_0 = \pm 1$.
d) Draw the qualitative phase line.

2.5 ★★ Find the fixed points of $x_{n+1} = \sqrt{x_n}$ (assuming $x_n \geq 0$) and determine their stability. What is the long-term behavior for any $x_0 > 0$?

2.6 ★ For the logistic map $x_{n+1} = r\,x_n(1 - x_n)$, describe qualitatively the change in long-term behavior as r increases from 0 to 4. What is the role of the Feigenbaum constant in this cascade?

2.7 Starting from $x_0 = 0.2$, iterate the logistic map $x_{n+1} = 2.5\,x_n(1 - x_n)$ for three steps by hand. What does the long-term behavior look like, and why?

2.8 ★★★ A population is divided into three stages: larvae (L), young adults (Y), and adults (A). Suppose 1% of larvae become young adults each year, 30% of young adults become adults. Young adults produce 104 offspring/year; adults produce 160 offspring/year. The yearly death rates are 3%, 30%, and 50% for L, Y, A respectively.

a) Write a system of difference equations modeling the dynamics.
b) Starting with $L_0 = 100$, $Y_0 = 0$, $A_0 = 0$, compute the state after two years.

2.9 ★★ On an island the population is divided into *lovers* and *haters* of dolphins. Every day each person talks to every other person exactly once. There is a fixed probability that a lover becomes a hater and vice versa. Using difference equations, model the time evolution of each group. State your assumptions and discuss the long-term dynamics.

2.10 ★ What is a *bifurcation diagram*? Sketch (conceptually, or describe) the bifurcation diagram of the logistic map and identify the onset of the first period-doubling and the onset of chaos.

2.11 ★★ Comment: *"Non-autonomous, higher-order difference equations can always be converted into autonomous, first-order forms."*

2.12 ★★ Comment: *"All first-order discrete linear systems can be solved algebraically."*

3 Continuous Dynamical Systems

*Continuous systems evolve according to ODEs $\dot{x} = f(x)$. **Fixed points** satisfy $f(x^*) = 0$ and are stable when $f'(x^*) < 0$ (attractors) and unstable when $f'(x^*) > 0$ (repellers). The **phase line** (1D) or **phase plane** (2D) visualise trajectories. Key 1D fact: a first-order autonomous continuous system cannot oscillate. In 2D we get richer behavior: stable spirals, limit cycles, saddle points. Numerical integration methods include **Euler** ($O(\Delta t)$ error) and **Runge-Kutta 4** ($O((\Delta t)^4)$ error). The **Lotka-Volterra** and **Lorenz** equations are central case studies, as is the **Romeo & Juliet** model.*

3.1 ★ Consider $\dot{x} = 2x - x^2$.

a) Find all fixed points.
b) Sketch the phase line with arrows and classify each fixed point (attractor, repeller, or half-stable).
c) What is the long-term behavior for initial conditions $x_0 > 0$?

3.2 ★★ Consider $\dot{x} = \sin(x)$.

a) This is a first-order autonomous continuous system. Comment: does it exhibit oscillatory behavior?
b) Find the fixed points and classify them.

3.3 ★★★ Comment: *"The dynamical system $\dot{x} = f(x)$ cannot exhibit oscillatory behavior whatever form $f(x)$ may take."*

3.4 ★ Comment: *"We only need to study linear first-order autonomous (LFA) systems because all other types of systems can be converted into an LFA system."*

3.5 ★★ You make a soup and it's very hot. As a dynamical systems scientist you decide to model the cooling process with a differential equation. Write down a physically motivated model, state your assumptions, draw the phase space, and identify the fixed points and their nature.

3.6 ★★ Explain the **Euler method** for numerically solving $\dot{x} = f(x)$. Simulate the first two steps for $\dot{x} = x^2 - 4$, starting from $x_0 = 3$, using step size $\Delta t = 0.1$.

3.7 ★ What are the key differences between the Euler method and the fourth-order Runge-Kutta method in terms of accuracy and computational cost? When would you choose one over the other?

3.8 ★★ Present the **Lotka-Volterra** predator-prey model. Identify all fixed points and describe the qualitative behavior of the system (oscillations, nullclines, phase portrait).

3.9 ★★ In class we studied the **Romeo & Juliet** model of a love affair. The model has two variables and four parameters. Write the general model and choose parameter values to describe a *cautious lover*: one who is encouraged by the other's positive feelings but retreats from their own feelings.

3.10 ★ The mid-semester exam considered a model of supply and demand: the price rises proportionally when quantity is below a threshold, and falls when above; quantity increases proportionally when price is above its threshold, and decreases below. Write and justify differential equations for this system. How does it relate to the Lotka-Volterra model?

3.11 ★ An ecosystem has two competing predators (e.g., lions and tigers) and one prey (e.g., zebras). Extend the standard Lotka-Volterra equations to cover this scenario. Justify each new term you introduce.

3.12 ★★ Two species (A and B) live in **symbiosis**: both can survive alone, but together they thrive. Model this cooperative ecosystem using differential equations. Justify each term, find the fixed points, and describe the qualitative dynamics.

3.13 ★★ Consider the system $\dot{x} = x$. Can this system exhibit chaotic behavior? Justify rigorously.

4 Chaos

*A system is **chaotic** if it is (1) deterministic, (2) aperiodic (bounded but non-repeating), (3) bounded, and (4) exhibits **sensitive dependence on initial conditions (SDIC)** — two arbitrarily close trajectories diverge exponentially. The **Lyapunov exponent** λ quantifies this: $\lambda > 0$ means chaos. Chaos does* not *mean random: it is fully deterministic, and its long-run statistics are often stable (ergodicity). The **Lorenz system** and the logistic map in the $r > 3.57$ regime are canonical examples. The bifurcation cascade follows the universal **Feigenbaum constant** $\delta \approx 4.669$.*

4.1 ★★ *"A chaotic system is different from other types of dynamical systems due to the random nature of its behavior."* Comment on this statement.

4.2 ★★★ State the four defining conditions of chaos. For each condition, explain its meaning and why it is necessary (i.e., why removing that condition alone would not give chaos).

4.3 ★★ *"The logistic equation in the chaotic regime is statistically predictable even if individual orbits are unpredictable."* Comment.

4.4 ★ Define the **Lyapunov exponent** λ. What does $\lambda > 0$ imply? What does $\lambda = 0$ typically signal in the context of the logistic map's bifurcation diagram?

4.5 ★ What is a **strange attractor**? How does it differ from a fixed-point attractor or a limit cycle? Give an example from the course.

4.6 Describe the **Lorenz system**: write the equations, name the parameters used in class, and explain why it is considered a landmark result in chaos theory.

4.7 ★ What is **ergodicity** in the context of chaos? Why is it relevant for the practical use of chaotic models in science?

4.8 ★★ Define the **Feigenbaum constant** δ. Why is it called "universal"?

4.9 ★ Briefly distinguish *period-2 orbit*, *quasiperiodic orbit*, and *chaotic orbit* in a discrete system. How can you distinguish them in a bifurcation diagram vs. a time series?

4.10 ★★ Comment: *"There is a relationship between the bifurcation diagram and the Lyapunov exponent."* Describe this relationship precisely.

4.11 ★★ Define the **Chaos Game**. What does it produce and why does it work?

5 Fractals

*A **fractal** is an object with **self-similarity** across scales. Its **self-similarity dimension** $D = \log N_C / \log M_F$ exceeds its **topological dimension** (e.g., a curve has topological dim. 1, but can have $D > 1$). Fractals can be generated via **Iterated Function Systems (IFS)**: either deterministically (DIFS) or randomly (RIFS, the "Chaos Game"). The key theoretical result is the **Contraction Mapping Principle** (Banach): a complete metric space with a contractive IFS has a unique fixed-point set (the **attractor**). Classic examples: Cantor set, Sierpiński triangle, Koch curve.*

5.1 ★★ *"A fractal object has a self-similarity dimension greater than its topological dimension."* Comment. Is this always the case?

5.2 ★ Define **self-similarity dimension**. Compute the self-similarity dimension of:

a) The Cantor set (remove the middle third at each step).
b) The Sierpiński triangle.
c) The Koch curve (each segment replaced by 4 segments of length 1/3).

5.3 ★ Explain the difference between a **deterministic IFS (DIFS)** and a **random IFS (RIFS)**. Both can produce the same attractor — how is the RIFS (Chaos Game) approach justified?

5.4 State the **Contraction Mapping Principle** (Banach Fixed-Point Theorem) as applied to IFS. What are the conditions, and what does it guarantee about the attractor?

5.5 ★ Describe the **Chaos Game** procedure for drawing the Sierpiński triangle. Why does it work, even though the starting point is chosen at random?

5.6 ★ Is the topological dimension of a fractal always an integer? Is the self-similarity dimension? What are their typical values for curves and surfaces?

5.7 Give an informal argument for why the coastline of Britain is considered a fractal. What does "measuring" the coastline with rulers of different lengths reveal, and how does this connect to the concept of fractal dimension?

5.8 ★★ The Sierpiński triangle was discussed in class. Describe **all three** methods studied for obtaining it, and identify the key conceptual idea behind each approach.

5.9 Give two concrete applications of fractal geometry (e.g., in engineering, computer graphics, or biology) discussed in class. For each, explain which fractal property is being exploited.

6 Julia & Mandelbrot Sets

Note: *Julia and Mandelbrot sets appear in past exams but are not explicitly covered in the current year's (`t/`) slides — they were a separate module in previous years. Review this section if your professor has covered it; otherwise treat it as supplementary.*
Julia and Mandelbrot sets arise from iterating $f_c(z) = z^2 + c$ in the complex plane. The ***filled Julia set*** *K_c for a fixed c is the set of initial values z_0 whose orbits remain bounded. Its boundary is the* ***Julia set*** *J_c. The* ***Mandelbrot set*** *$\mathcal{M}$ is the set of c values for which the orbit of $z_0 = 0$ remains bounded — it is a "map" of Julia sets. The* ***Julia–Fatou theorem*** *links J_c's connectivity to whether $c \in \mathcal{M}$. Julia sets are also* ***attractors of an IFS****, connecting fractals and complex iteration.*

6.1 ★★ Define the **Julia set** J_c for $f_c(z) = z^2 + c$. What is the filled Julia set K_c? What is the role of the *critical orbit* (the orbit of $z_0 = 0$)?

6.2 ★★ How can one obtain a Julia set using an **Iterated Function System (IFS)**? Describe the procedure, including the required inverse functions.

6.3 ★★ Define the **Mandelbrot set** $\mathcal{M}$. How does it relate to the family of Julia sets $\{J_c\}_{c\in\mathbb{C}}$?

6.4 State the **Julia-Fatou theorem** connecting the shape of J_c to the membership of c in $\mathcal{M}$. What happens to J_c when c is outside $\mathcal{M}$?

6.5 ★ Explain why Julia sets are considered fractals. Which fractal property (self-similarity, non-integer dimension, etc.) is most directly visible in their structure?

7 Network Science

*A **graph** $G = (V, E)$ models pairwise relations. Key metrics: **degree**, **clustering coefficient** (local density of triangles), **diameter** (longest shortest path), **density** (actual vs. possible edges). **Centrality measures** rank nodes by importance: degree, closeness, betweenness. Two famous random graph models: **Erdős-Rényi** (random edges, Poisson degree dist.) and **Barabási-Albert** (preferential attachment, power-law "scale-free" degree dist.). **Small-world** networks (Watts-Strogatz) combine high clustering with short paths. **Community detection** (Girvan-Newman, Louvain) partitions the graph by maximising **modularity** Q. **Dynamics on networks**: voter model, SIS epidemic, diffusion, Kuramoto synchronisation.*

7.1 ★★★ Define the **diameter** of a graph and the **radius** of a graph. What is the *eccentricity* of a node? Give the relationships between diameter, radius, and eccentricity, and identify the *center* and *periphery* of a graph.

7.2 ★★ Define the **clustering coefficient** of a node v. Compute it for a node with 4 neighbors among which there are 3 edges.

7.3 ★★ Define **bipartite graph**. Give an example. Can a bipartite graph have odd-length cycles?

7.4 Define the three centrality measures studied in class (degree, closeness, betweenness). For each, give an example of a real-world situation where it is the most informative measure.

7.5 ★ Describe the **Barabási-Albert** model. What mechanism produces the scale-free degree distribution? Write the approximate power-law form of the degree distribution.

7.6 ★★ What is a **small-world network**? Describe the Watts-Strogatz construction procedure. What two structural properties make it "small-world"?

7.7 ★★★ *"In terms of robustness it is better to have a system modeled as a small-world network than one whose model is a scale-free network."* Comment.

7.8 Define **modularity** Q. What does $Q \approx 0$ mean and what does a large Q mean? Outline the Girvan-Newman community detection algorithm.

7.9 ★ Describe the **voter model** as a dynamics-on-networks model. What is the absorbing state and under what conditions does the system reach it?

7.10 Describe the **SIS model** (Susceptible-Infected-Susceptible) on a network. How does the network structure affect the epidemic threshold compared to a well-mixed population?

7.11 What is the **Kuramoto model**? What phenomenon does it describe, and what is the key order parameter? Give a real-world example of the phenomenon it models.

7.12 Define **assortativity** in a network. What is *degree assortativity*? Give an example of a real-world network that is positively assortative and one that is negatively assortative.

7.13 Define **network percolation** and **giant component**. In an Erdős-Rényi random graph $G(N, p)$, at what critical condition does a giant component appear? What does this tell us about the robustness of the network?

7.14 Describe the **diffusion model** on a network. Write the update equation for node i's concentration and explain the role of the network topology in determining how quickly quantities spread.

8 Agent-Based Models & Cellular Automata

A ***cellular automaton (CA)*** *is a tuple* (S, s_0, N, d, f)*: a state set* S*, initial config* s_0*, neighborhood* N*, dimension* d*, and local rule* f*. Wolfram classified 1D binary CAs into 4 classes (fixed, periodic, complex/fractal, chaotic). The number of distinct rules for a binary CA of radius* r *is* $2^{2^{2r+1}}$*.* ***Conway's Game of Life*** *is a famous 2D CA with rich emergent behavior.* ***Agent-based models*** *generalise CAs: agents have state, sense their environment, and act according to rules. The* ***Schelling segregation model*** *shows how mild individual preferences lead to strong global segregation (emergence).*

8.1 ★★ Define **cellular automaton** formally (state all five components of the tuple). Describe the difference between a **Moore neighborhood** and a **Von Neumann neighborhood** in 2D.

8.2 ★★★ *"In agent-based modeling a Moore neighborhood is preferable to a Von Neumann neighborhood."* Comment.

8.3 How many distinct rules are there for a binary CA ($q = 2$) with radius $r = 1$ in 1D? Show the calculation. What about $r = 2$?

8.4 ★ Describe **Wolfram's four classes** of elementary CA behavior. Give the key observable feature of each class (e.g., what you see on a space-time diagram).

8.5 ★★ Define **Game of Life**. State the four rules governing cell birth and death. Give an example of a static configuration, a periodic configuration, and a "glider."

8.6 ★★ Define the **Schelling segregation model**. What does the model demonstrate about the relationship between individual preferences and collective outcomes? What concept does this illustrate?

8.7 ★ What are the key differences between agent-based models and differential equation models? Under what circumstances might an ABM reveal dynamics that an ODE model would miss?

8.8 ★★★ In class we studied a predator-prey system modeled three ways: difference equations, differential equations, and agent-based. Compare the ABM approach and the network-based approach for predator-prey. In what way is each more appropriate?

8.9 ★ Explain the concept of **self-organisation** using a cellular automaton example from the course. Why is no external controller needed to produce ordered patterns?

8.10 ★★ Consider a 1D cellular automaton with $q = 3$ states $(\oplus, \otimes, \odot)$ and radius $r = 1$ but where the transition of each cell depends *only on its two immediate neighbors* (not itself). How many distinct transition rules exist? Show the calculation.

8.11 ★★★ Someone claims to have observed the following state transition in a 1D binary CA with radius $r = 1$:

$$t: \quad 0,1,0,0,1,0,1,1,1,0,0 \qquad t+1: \quad 0,1,1,0,0,1,0,1,0,1,0$$

Is this transition consistent with a single valid CA rule? Identify all neighborhood triples that appear, check for contradictions, and determine (or reconstruct) the rule.

8.12 What is **majority rule** dynamics on a network?

8.13 ★ Describe an **agent-based version of the SIR model** (Susceptible–Infected–Recovered). Specify: what each agent's state is, what rules govern state transitions, how agents interact, and what advantages this approach has over the differential-equation SIR model.

8.14 ★ Starting from the agent-based SIR model in the previous question, suppose agents can now **travel by plane** between distant cities. Describe how you would extend the model to incorporate this. What new dynamics or phenomena does this extension enable? How does it differ from the voter model? What is the absorbing state in each case?

9 Graph Neural Networks

Graph Neural Networks (GNNs) *extend deep learning to graph-structured data. The key mechanism is* ***message passing****: each node iteratively aggregates representations from its neighbors to build a richer embedding.* ***GCN*** *(Graph Convolutional Network) is transductive — it requires the full graph at training time.* ***GraphSAGE*** *is inductive — it learns an aggregation function and can generalise to unseen nodes/graphs. Tasks:* ***node classification****,* ***link prediction****,* ***graph classification****. GNNs sit at the intersection of complex systems and ML: they can be used both to analyze networks and to model dynamics on them.*

9.1 Describe the **message-passing** framework of GNNs. What are the three steps in each layer and what does the node embedding represent after k layers?

9.2 What is the difference between **transductive** and **inductive** learning in the context of GNNs? Give one example of a task that requires inductive learning and one that can be handled transductively.

9.3 Compare **GCN** and **GraphSAGE**: how does each aggregate neighbor information, and what is the practical implication of the difference?

9.4 Name and briefly describe three different prediction tasks that GNNs can perform, giving a concrete application for each.

Answer Key

Section 1 — Concepts & Models

1.1 A **complex system** is a system of many interacting components whose collective behavior is *qualitatively different* from the behavior of its parts: it exhibits emergence, nonlinearity, feedback, self-organisation, and often adaptation. A "complicated" system (e.g., a jet engine) can in principle be fully understood by disassembling it; a complex system (e.g., ant colony, economy) cannot — the interesting properties live in the interactions, not the parts.

1.2 **Emergence** is the appearance of global properties or behaviors in a system that are not present in, nor predictable from, the individual components alone. Example: Conway's Game of Life — gliders and stable structures emerge from four simple local rules applied to binary cells.

1.3

- **Mathematical models (difference/differential equations)**: state = continuous or discrete variable(s); time = discrete steps or continuous; example = logistic map, Lotka-Volterra.
- **Agent-based models**: state = collection of agent attributes; time = discrete steps; example = Schelling segregation, Game of Life.
- **Network models**: state = graph $G = (V, E)$ plus node/edge attributes; dynamics modeled via rules on the graph; example = SIS epidemic, voter model.

1.4 **Mathematical**: analytically tractable, easy to fit parameters, reveals equilibria and stability; but assumes spatial homogeneity (well-mixed) and ignores individual variation. **ABM**: captures heterogeneity, spatial structure, and emergent behavior; but parameter-rich, harder to analyze theoretically. **Network**: captures contact structure and propagation paths; but defining the network and updating it is costly. Yes, these trade-offs generalise: mathematics gives global insight at the cost of abstraction; ABM/network add realism at the cost of complexity.

1.5 **Deterministic**: given the same initial condition, the evolution is unique (e.g., logistic map, Lorenz). **Stochastic**: evolution involves random elements — e.g., RIFS (random IFS), SIS model with probabilistic infection, voter model. Both can coexist: even deterministic systems like the logistic map in the chaotic regime are *practically* indistinguishable from random because of SDIC.

1.6

a) Bank account: **discrete, linear, autonomous, order 1** (first-order: only x_t appears, no external forcing).
b) Logistic map: **discrete, nonlinear** (due to x_t^2 term), **autonomous, order 1**.
c) Lotka-Volterra: **continuous, nonlinear** (xy product), **autonomous, order 2** (two state variables).
d) Lorenz: **continuous, nonlinear** (xz, xy products), **autonomous, order 3**.

1.7 Boids rules: (1) **Separation** — avoid crowding neighbors; (2) **Alignment** — steer toward the average heading of neighbors; (3) **Cohesion** — steer toward the average position of neighbors. From these purely local rules, global coherent flocking patterns emerge with no central controller. This illustrates **emergence** and **self-organisation**: complex collective behavior arising from simple local interactions.

Section 2 — Discrete Dynamical Systems

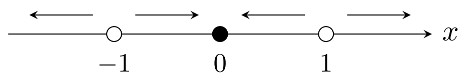
2.1 A **fixed point** x^* of f satisfies $x^* = f(x^*)$. Graphically, it is the intersection of $y = f(x)$ with the diagonal $y = x$. On a cobweb diagram, the orbit "lands" exactly on x^* and stays there.

2.2 A **cobweb diagram** is drawn by starting at (x_0, x_0), moving vertically to $(x_0, f(x_0))$, then horizontally to $(f(x_0), f(x_0))$, and repeating. If the cobweb spirals *inward* toward a fixed point it is stable; if it spirals *outward* it is unstable.

2.3 The **basin of attraction** of a stable fixed point x^* is the set of initial conditions x_0 from which the orbit converges to x^*. Not every fixed point has one: unstable fixed points do not (they repel all nearby trajectories).

2.4 $f(x) = x^3$.

a) Fixed points: $x^* = (x^*)^3 \Rightarrow x^*(x^{*2} - 1) = 0$, so $x^* \in \{-1, 0, 1\}$.
b) $f'(x) = 3x^2$. At $x^* = 0$: $|f'(0)| = 0 < 1$ **stable**. At $x^* = \pm 1$: $|f'(\pm 1)| = 3 > 1$ **unstable**.
c) For $|x_0| < 1$: orbit $\to 0$. For $|x_0| > 1$: orbit $\to \pm\infty$. For $x_0 = \pm 1$: fixed (but any tiny perturbation escapes).
d) Phase line ($\circ$ = unstable, $\bullet$ = stable):



2.5 $f(x) = \sqrt{x}$. Fixed points: $x^* = \sqrt{x^*} \Rightarrow x^* \in \{0, 1\}$. $f'(x) = \frac{1}{2\sqrt{x}}$. At $x^* = 0$: $f'(0) \to \infty > 1$ **unstable**. At $x^* = 1$: $f'(1) = \frac{1}{2} < 1$ **stable**. For any $x_0 > 0$, the orbit converges to 1.

2.6 As r increases from 0 to 4: $r < 1$ all orbits $\to 0$ (extinction); $1 < r < 3$ stable non-zero FP; $r = 3$ first bifurcation (period-2); period-doubling cascade; $r \approx 3.57$ onset of chaos. The **Feigenbaum constant** $\delta \approx 4.669$ is the ratio of successive bifurcation intervals — it is *universal*: the same constant appears in any smooth unimodal map.

2.7 $r = 2.5$, $x_0 = 0.2$: $x_1 = 2.5(0.2)(0.8) = 0.4$; $x_2 = 2.5(0.4)(0.6) = 0.6$; $x_3 = 2.5(0.6)(0.4) = 0.6$. Since $1 < r = 2.5 < 3$, the system has a stable fixed point; orbits converge to $x^* = 1 - 1/r = 0.6$.

2.8

a) Let $d_L = 0.03, d_Y = 0.30, d_A = 0.50$; larvae mature at rate 0.01; YA to A at 0.30; offspring: 104 per YA, 160 per A.

$$L_{n+1} = (1 - d_L - 0.01)L_n + 104\,Y_n + 160\,A_n = 0.96\,L_n + 104\,Y_n + 160\,A_n$$

$$Y_{n+1} = 0.01\,L_n + (1 - d_Y - 0.30)Y_n = 0.01\,L_n + 0.40\,Y_n$$

$$A_{n+1} = 0.30\,Y_n + (1 - d_A)A_n = 0.30\,Y_n + 0.50\,A_n$$

b) Year 0: $(100, 0, 0)$. Year 1: $L_1 = 96$, $Y_1 = 1$, $A_1 = 0$. Year 2: $L_2 = 0.96(96) + 104(1) + 160(0) = 92.16 + 104 = 196.16$; $Y_2 = 0.01(96) + 0.40(1) = 0.96 + 0.40 = 1.36$; $A_2 = 0.30(1) + 0.50(0) = 0.30$.

2.9 Let L_n = lovers, H_n = haters, $N = L_n + H_n$ = const. Each person meets $N - 1$ others. Let p = probability a lover becomes a hater per meeting, q = vice versa. Per day: $L_{n+1} = L_n - p\,L_n(N-1)/N \cdot (\ldots) + q\,H_n(\ldots)$. A simple linear model: $L_{n+1} = (1-p)L_n + q\,H_n$; $H_{n+1} = p\,L_n + (1-q)H_n$ (with p, q lumped per-day transition rates). The system has a single stable fixed point at $L^* = \frac{q}{p+q}N$, regardless of initial conditions (assuming $p, q > 0$). Long-term: mixed coexistence.

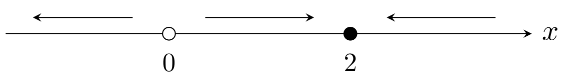
2.10 A **bifurcation diagram** plots the long-term values (attractor) of x_n as a function of the parameter r. For the logistic map: single line (stable FP) for $1 < r < 3$; first split at $r = 3$ (period-2); successive splits at decreasing intervals; a dense "chaotic" band for $r \gtrsim 3.57$ interspersed with periodic windows (e.g., period-3 near $r \approx 3.83$).

2.11 **True**. *Higher-order → first-order*: a kth-order equation $x_t = F(x_{t-1}, x_{t-2}, \ldots, x_{t-k})$ becomes first-order by introducing auxiliary variables, e.g. $x_t = x_{t-1} + x_{t-2}$ becomes $x_t = x_{t-1} + y_{t-1}$, $y_t = x_{t-1}$. *Non-autonomous → autonomous*: a time-dependent equation $x_t = F(x_{t-1}, t)$ becomes autonomous by adding a clock variable $z_t = z_{t-1} + 1$ (with $z_0 = 1$) so that t is replaced by z_{t-1}. Both conversions are always possible, so the statement is **true**.

2.12 **True**. A first-order linear discrete system has the form $x_{t+1} = a\,x_t + b$. This is a linear recurrence with closed-form solution $x_t = a^t x_0 + b\frac{1-a^t}{1-a}$ (for $a \neq 1$) or $x_t = x_0 + bt$ (for $a = 1$). More generally, any linear recurrence with constant coefficients can be solved algebraically via characteristic equations. The statement is **true**.

Section 3 — Continuous Dynamical Systems

3.1 $\dot{x} = 2x - x^2 = x(2 - x)$. FPs: $x^* = 0$ ($f'(0) = 2 > 0$ **repeller**) and $x^* = 2$ ($f'(2) = -2 < 0$ **attractor**). For $x_0 > 0$: orbit $\to 2$.

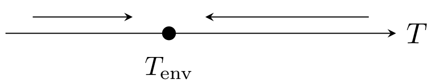


3.2 $\dot{x} = \sin(x)$. FPs where $\sin(x^*) = 0$: $x^* = k\pi$ for $k \in \mathbb{Z}$. $f'(x^*) = \cos(k\pi) = (-1)^k$. Odd k: $f' < 0$ stable; even k: $f' > 0$ unstable. **No oscillation**: a 1D autonomous continuous system cannot oscillate (trajectories cannot cross, and the phase line only allows monotone motion between FPs).

3.3 **True** for 1D autonomous systems: $\dot{x} = f(x)$ with $x \in \mathbb{R}$ has a monotone phase line; trajectories can only increase or decrease, never reverse without crossing a fixed point. **False** for higher-dimensional systems: $\dot{\mathbf{x}} = f(\mathbf{x})$ with $\mathbf{x} \in \mathbb{R}^n$, $n \geq 2$ can exhibit limit cycles (e.g., Lotka-Volterra).

3.4 **False**. Not all systems are reducible to LFA without loss: nonlinear systems can exhibit behavior (bifurcations, chaos, limit cycles) that no LFA can. Linearization around a fixed point is valid *locally* for small perturbations but cannot capture global dynamics. A system like $\dot{x} = x^2$ cannot be converted to a global LFA without losing its blow-up behavior.

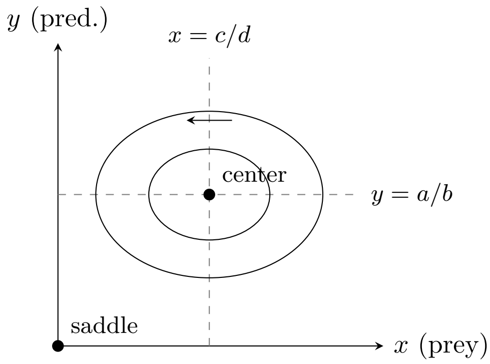
3.5 Newton's law of cooling: $\dot{T} = -k(T - T_{\text{env}})$, where T is soup temperature, T_{env} is room temperature (fixed), $k > 0$ is a cooling constant. Assumption: heat loss is proportional to the temperature difference; environment has infinite heat capacity. FP: $T^* = T_{\text{env}}$, stable ($f' = -k < 0$). All initial conditions converge to T_{env}.



3.6 Euler: $x_{t+\Delta t} = x_t + \Delta t \cdot f(x_t)$. For $\dot{x} = x^2 - 4$, $x_0 = 3$, $\Delta t = 0.1$: Step 1: $f(3) = 9 - 4 = 5$; $x_1 = 3 + 0.1 \cdot 5 = 3.5$. Step 2: $f(3.5) = 12.25 - 4 = 8.25$; $x_2 = 3.5 + 0.1 \cdot 8.25 = 4.325$. (Note: the true fixed points are ± 2; $x_0 = 3 > 2$ so the orbit diverges.)

3.7 Euler: first-order, $O(\Delta t)$ local error — simple but requires small Δt for accuracy. RK4: fourth-order, $O((\Delta t)^4)$ error — 4 function evaluations per step, far more accurate for smooth f. Choose Euler for quick prototypes or when f is cheap; RK4 for production simulations requiring accuracy.

3.8 Lotka-Volterra: $\dot{x} = ax - bxy$, $\dot{y} = -cy + dxy$. FPs: $(0, 0)$ (saddle) and $(c/d,\ a/b)$ (center). Nullclines: $y = a/b$ (prey, horizontal) and $x = c/d$ (predator, vertical). Closed CCW orbits around the center; amplitude depends on initial conditions.



3.9 Romeo-Juliet: $\dot{R} = a\,R + b\,J$, $\dot{J} = c\,R + d\,J$. Cautious lover (say, Romeo): retreats from his own feelings ($a < 0$) but responds positively to Juliet's feelings ($b > 0$). Example: $a = -0.2$, $b = 0.8$, $c = 0.5$, $d = 0.1$. The eigenvalues of the system matrix determine whether they reach a fixed point (stable love), oscillate, or diverge.

3.10 Let Q = quantity, P = price, Q_0, P_0 = thresholds. $\dot{P} = \alpha(Q_0 - Q)$; $\dot{Q} = \beta(P - P_0)$. This is a 2D linear system structurally identical to Lotka-Volterra (linearised around the interior fixed point), producing oscillatory behavior around (Q_0, P_0).

3.11 Let x = zebras (prey), y = lions, z = tigers. Both predators feed on x and compete with each other (no direct interaction term needed if they only compete for the same prey). One formulation:

$$\dot{x} = ax - b_1xy - b_2xz, \quad \dot{y} = -c_1y + d_1xy, \quad \dot{z} = -c_2z + d_2xz.$$

Terms $-b_1xy$, $-b_2xz$ reduce prey due to each predator; d_1xy, d_2xz grow each predator from prey consumption; $-c_1y, -c_2z$ = natural predator mortality. (Extensions: direct competition terms $-e_1yz$ in $\dot{y}$ and $+e_2yz$ in $\dot{z}$ if one dominates the other.)

3.12 Symbiosis model: let $x, y \geq 0$ be the populations of species A and B.

$$\dot{x} = r_1x - ax^2 + bxy \qquad \dot{y} = r_2y - cy^2 + dxy$$

Terms: r_1x = intrinsic growth (can survive alone); $-ax^2$ = intraspecific competition (carrying capacity); $+bxy$ = benefit from B's presence (cooperation). Symmetrically for y. The mutualistic terms $+bxy$ mean each species grows faster in the presence of the other. Fixed points include $(0, 0)$, $(r_1/a, 0)$, $(0, r_2/c)$, and an interior coexistence point. For strong cooperation ($bd > ac$), populations can grow without bound — a key difference from competition models.

3.13 **No**. $\dot{x} = x$ is a one-dimensional, first-order autonomous continuous system. For a 1D continuous system three conditions for chaos are violated: (a) **No SDIC possible**: solutions are $x(t) = x_0e^t$; two solutions with initial conditions x_0, $x_0 + \epsilon$ stay separated by ϵe^t — they diverge, but this is not SDIC in the bounded sense. More fundamentally, (b) **chaos requires** $n \geq 3$ **dimensions** for continuous autonomous systems (Poincaré-Bendixson: in $\mathbb{R}^2$ bounded trajectories converge to FPs or limit cycles; chaos needs $\mathbb{R}^3$ or higher). This 1D system simply diverges exponentially — that is not chaos (chaos requires boundedness).

Section 4 — Chaos

4.1 **False**. Chaos is *deterministic*: given the same initial conditions the trajectory is identical. What makes it *appear* random is **sensitive dependence on initial conditions (SDIC)**: arbitrarily close initial states diverge exponentially, so the long-term orbit is practically unpredictable without infinite precision. Chaos is fully causal; it is not random.

4.2 (1) **Deterministic**: the rule is fixed, no stochasticity. (2) **Aperiodic**: the orbit never exactly repeats (otherwise predictability would return). (3) **Bounded**: the orbit stays in a finite region (otherwise it would just diverge, which is trivial). (4) **SDIC**: nearby orbits diverge exponentially — this is the operational signature of chaos, captured by $\lambda > 0$.

4.3 **True (partially)**. Individual orbits are unpredictable due to SDIC. However, the **invariant measure** of the chaotic attractor is stable: time-averaged statistics (e.g., mean, variance, histogram of visited values) converge to fixed values for almost all initial conditions (**ergodicity**). This makes long-run ensemble statistics predictable even when individual trajectories are not — e.g., climate vs. weather.

4.4 $\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)|$. Measures average exponential divergence rate. $\lambda > 0$: chaos (SDIC). $\lambda < 0$: stable attractor. $\lambda = 0$: marginal — occurs *at bifurcation points* in the logistic map's bifurcation diagram, marking the transition between periodic and chaotic regimes.

4.5 A **strange attractor** is a bounded, aperiodic attractor with a fractal structure. Unlike a fixed point (0D) or limit cycle (1D), it has a non-integer Hausdorff dimension and nearby trajectories on it diverge (SDIC). Example: the **Lorenz attractor** — butterfly-shaped, $D \approx 2.06$.

4.6 Lorenz: $\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - bz$. Classic parameters: $\sigma = 10$, $b = 8/3$, $\rho = 28$. Originally a simplified atmospheric convection model. Landmark result: first clear demonstration that a *simple 3D deterministic system* can produce aperiodic, unpredictable behavior — the origin of the "butterfly effect" metaphor.

4.7 **Ergodicity**: a chaotic system visits all parts of its attractor with a stationary long-run frequency — time averages equal ensemble averages. Practical implication: even though individual predictions fail, *statistical* predictions (e.g., what fraction of time the system spends above a threshold) are well-defined and stable.

4.8 The **Feigenbaum constant** $\delta \approx 4.669$ is the universal limiting ratio of successive bifurcation-interval lengths in the period-doubling cascade: $\delta = \lim_{n\to\infty}(r_n - r_{n-1})/(r_{n+1} - r_n)$. *Universal* means: it is the same for *any* smooth unimodal map (quadratic, sin, cubic, etc.) — a striking result connecting chaos to renormalisation theory.

4.9 **Period-2 orbit**: the sequence repeats every 2 steps (e.g., $\ldots a, b, a, b, \ldots$) — on a bifurcation diagram, two branches; on a time series, regular alternation. **Quasiperiodic**: two incommensurate frequencies — time series looks almost periodic but never exactly repeats; power spectrum has two sharp peaks. **Chaotic**: no periodicity; broadband power spectrum; positive Lyapunov exponent; on bifurcation diagram, a dense cloud of points.

4.10 **True, and it is a precise and important relationship**. The Lyapunov exponent $\lambda(r)$ is computed for each parameter value r. In the bifurcation diagram: (i) at each **period-doubling bifurcation**, $\lambda = 0$ (the boundary between stable and unstable); (ii) in **stable periodic windows**, $\lambda < 0$ (fixed-point or periodic attractor); (iii) in the **chaotic band**, $\lambda > 0$; (iv) the **onset of chaos** ($r \approx 3.57$) corresponds to $\lambda = 0$ permanently. So the Lyapunov exponent can be read as a "detector" of the structure seen in the bifurcation diagram.

4.11 The **Chaos Game** (RIFS): (1) choose any starting point; (2) at each step, randomly select one of the IFS contractions (with assigned probabilities) and apply it to the current point; (3) plot the result. After a short transient, all plotted points lie on the attractor of the IFS. It works because the attractor is the unique fixed point of the IFS operator (Banach theorem) and the random orbit visits all parts of it according to the natural invariant measure.

Section 5 — Fractals

5.1 True — and it is the definition. The course defines a fractal as a geometric object whose self-similarity dimension D is greater than its topological dimension d_T (slide 20). So the statement is not merely a typical property: it is what makes something a fractal. Examples: Cantor set ($d_T = 0$, $D \approx 0.63$); Sierpiński triangle ($d_T = 1$, $D \approx 1.585$); Koch curve ($d_T = 1$, $D \approx 1.262$). Note: the dimension D need not be non-integer — a space-filling curve has $d_T = 1$ and $D = 2$ (an integer), but $D > d_T$ still holds, so it is a fractal by this definition.

5.2 $D = \log N_C / \log M_F$.

a) Cantor: each step removes middle third, leaving $N_C = 2$ pieces of scale $M_F = 3$. $D = \log 2/\log 3 \approx 0.631$.
b) Sierpiński triangle: $N_C = 3$ copies at $M_F = 2$. $D = \log 3/\log 2 \approx 1.585$.
c) Koch curve: $N_C = 4$ segments of length $1/3$, so $M_F = 3$. $D = \log 4/\log 3 \approx 1.262$.

5.3 DIFS: apply all transformations in sequence at every step (deterministic, entire attractor drawn at once). **RIFS**: at each step choose one transformation at random with assigned probability and plot the resulting point. Both converge to the same attractor by the Banach Fixed-Point Theorem: the attractor is the unique fixed point of the IFS operator in the complete metric space of compact sets. The RIFS produces points *distributed* over the attractor according to the natural invariant measure.

5.4 Contraction Mapping Principle: if (X, d) is a complete metric space and $T : X \to X$ is a contraction ($d(Tx, Ty) \le c\, d(x, y)$, $c < 1$), then T has a unique fixed point x^* and $T^n x \to x^*$ for all x. For IFS: the space is compact subsets of $\mathbb{R}^n$ with Hausdorff metric; the IFS operator $T(A) = \bigcup_i f_i(A)$ is contractive; the unique attractor A^* satisfies $T(A^*) = A^*$.

5.5 Choose any point in the triangle (or even outside). Repeatedly: pick a vertex at random with probability $1/3$; move to the midpoint between the current point and that vertex; plot. After a transient, all plotted points lie on the Sierpiński triangle. Justification: this is exactly a RIFS with three affine contractions mapping to the three sub-triangles; by the Banach theorem the orbit converges to the attractor.

5.6 Topological dimension is always a non-negative integer (it is a combinatorial/topological invariant). Self-similarity dimension need not be: for fractals it is typically a non-integer between the topological dimension and the embedding dimension. Curves: $d_T = 1$, $D \in (1, 2)$; surfaces: $d_T = 2$, $D \in (2, 3)$.

5.7 When measured with a ruler of length ℓ, the estimated coastline length $L(\ell) \sim \ell^{1-D}$ grows as $\ell \to 0$, with $D > 1$. Richardson (1961) found empirically that the British coastline has $D \approx 1.25$. This means the coastline has infinite length in the limit of infinite precision — it is not rectifiable — a hallmark of fractal geometry.

5.8 Three methods for the Sierpiński triangle:

a) **Geometric deletion (DIFS)**: Start with a filled equilateral triangle. Repeatedly remove the central upward triangle from each remaining filled triangle. After infinite iterations, the Cantor-like limiting set is the Sierpiński triangle. Conceptual idea: self-similar construction by deterministic subdivision.
b) **Iterated Function System — deterministic (DIFS)**: Define 3 affine contractions f_1, f_2, f_3 each scaling by $1/2$ toward a different vertex. Apply all three simultaneously at every step to the current set S: $S_{n+1} = f_1(S_n) \cup f_2(S_n) \cup f_3(S_n)$. Converges to the attractor by the Banach Fixed-Point Theorem. Conceptual idea: the triangle is the fixed point of a contractive operator on compact sets.
c) **Random IFS (RIFS / Chaos Game)**: Start at any point. At each step, randomly

choose one of the three contractions (probability 1/3 each) and apply it. Plot the result. After a transient, points trace out the Sierpiński triangle. Conceptual idea: the random orbit visits the attractor according to its invariant measure; randomness generates the same deterministic object.

5.9 (i) **Image compression via IFS**: an image can be encoded as a set of IFS contractions whose attractor approximates the image. Storing the (few) contraction parameters is vastly cheaper than storing every pixel — exploits the fact that natural images have approximate self-similarity. (ii) **Fractal antennas**: a fractal-shaped antenna (e.g., Koch-curve boundary) fits more perimeter into a compact area while maintaining multiband resonance — exploits high fractal dimension (large perimeter per area) and self-similarity (resonance at multiple scales).

Section 6 — Julia & Mandelbrot Sets

6.1 The **filled Julia set** $K_c = \{z_0 \in \mathbb{C} : |f_c^n(z_0)| \not\to \infty\}$. The **Julia set** $J_c = \partial K_c$ (boundary of K_c). The critical orbit is the orbit of $z_0 = 0$ (the critical point of f_c). If this orbit escapes to ∞, then J_c is a Cantor (totally disconnected) dust; if it stays bounded ($c \in \mathcal{M}$), J_c is connected.

6.2 To generate J_c via IFS: invert $f_c(z) = z^2 + c$ to get $f_c^{-1}(z) = \pm\sqrt{z - c}$ (two branches). Repeatedly apply a randomly chosen branch to a starting point $z_0 \in J_c$ (or any nearby point after a transient). The orbit traces out J_c because J_c is the repelling fixed-point set of the backward iteration. (This is the RIFS/Chaos Game applied to the complex map.)

6.3 The **Mandelbrot set** $\mathcal{M} = \{c \in \mathbb{C} : |f_c^n(0)| \not\to \infty\}$. It is the set of parameters c for which the critical orbit stays bounded. $\mathcal{M}$ is a "catalog" of Julia sets: if $c \in \mathcal{M}$, J_c is connected; if $c \notin \mathcal{M}$, J_c is a Cantor set. The boundary $\partial\mathcal{M}$ itself is a fractal.

6.4 **Julia-Fatou theorem**: J_c is either (i) connected (when the orbit of 0 is bounded, i.e., $c \in \mathcal{M}$) or (ii) a Cantor set (when $c \notin \mathcal{M}$). For c outside $\mathcal{M}$, $K_c = J_c$ (the filled Julia set has no interior), and it is totally disconnected.

6.5 Julia sets are fractals in the sense that (i) they exhibit self-similarity at all scales — zooming in reveals the same intricate structure — and (ii) they have a Hausdorff dimension strictly between 1 (a curve) and 2 (a filled region), for most values of c. The most visually apparent property is the infinite detail / self-similarity under magnification.

Section 7 — Network Science

7.1 The **eccentricity** of node u is $\varepsilon(u) = \max_{v\in V} d(u, v)$ — the greatest distance from u to any other node. The **diameter** is $\operatorname{diam}(G) = \max_u \varepsilon(u)$ — the largest eccentricity. The **radius** is $\operatorname{rad}(G) = \min_u \varepsilon(u)$ — the smallest eccentricity. Always $\operatorname{rad}(G) \le \operatorname{diam}(G) \le 2\operatorname{rad}(G)$. The **center** $C(G) = \{u : \varepsilon(u) = \operatorname{rad}(G)\}$ is the set of nodes with minimum eccentricity (closest to everyone). The **periphery** $P(G) = \{u : \varepsilon(u) = \operatorname{diam}(G)\}$ is the set of nodes farthest from everyone else.

7.2 $c_v = \frac{2e_v}{k_v(k_v-1)}$, where e_v is the number of edges among the k_v neighbors of v. With $k_v = 4$, $e_v = 3$: $c_v = 2 \cdot 3/(4 \cdot 3) = 6/12 = 0.5$.

7.3 A **bipartite graph** has vertices partitioned into two sets U, V such that every edge connects a U-node to a V-node. Example: student-course enrollment graph. **No**, a bipartite graph cannot have odd-length cycles — a graph is bipartite if and only if it has no odd-length cycles (standard graph theory result; not to be confused with König's theorem, which concerns maximum matchings).

7.4 **Degree centrality**: counts connections; best for identifying hubs/popular nodes (e.g., Twitter followers). **Closeness**: measures how quickly a node can reach all others; best for spread of information (e.g., disease spreader). **Betweenness**: counts how many shortest paths pass through a node; best for identifying bridges or bottlenecks (e.g., critical infrastructure nodes).

7.5 Barabási-Albert: start from a small graph; at each step add a new node with m edges connected to existing nodes with probability proportional to their degree (**preferential attachment** — "rich get richer"). This produces a **power-law** degree distribution: $P(k) \sim k^{-\gamma}$ with $\gamma \approx 3$. The network is **scale-free**: hubs (high-degree nodes) dominate.

7.6 A **small-world network** has (i) high clustering coefficient (like a regular lattice) and (ii) short average path length (like a random graph). Watts-Strogatz: start with a ring of n nodes each connected to k nearest neighbors; rewire each edge with probability p to a random node. At intermediate p: high clustering preserved, short paths introduced.

7.7 **Nuanced**. Scale-free networks are robust to **random failure** (removing a random node rarely hits a hub) but fragile to **targeted attack** (removing hubs rapidly disconnects the network). Small-world networks are more uniformly robust. So the statement is *partially true*: for targeted attacks, small-world is safer; for random failures, scale-free may be more robust.

7.8 Modularity $Q = \sum_c [L_c/m - (d_c/2m)^2]$. $Q \approx 0$: community structure is no better than random; large Q (> 0.3): strong community structure. Girvan-Newman: iteratively remove the edge with highest betweenness centrality (recomputed after each removal); the dendrogram reveals the hierarchical community structure. Stopping criterion: maximise Q.

7.9 **Voter model**: each node holds an opinion (e.g., ± 1). At each step a random node copies the opinion of a random neighbor. **Absorbing state**: consensus (all nodes agree). On finite networks consensus is always eventually reached; time to consensus depends on network structure (slower on clustered networks, faster on small-world/random).

7.10 SIS on a network: infected node i recovers at rate μ; susceptible node j connected to i gets infected at rate β. The epidemic threshold $\beta/\mu > 1/\lambda_{\max}$ (where $\lambda_{\max}$ is the largest eigenvalue of the adjacency matrix). For scale-free networks $\lambda_{\max} \sim \sqrt{k_{\max}}$ is large, so the threshold is *lower* — epidemics spread more easily than in homogeneous populations.

7.11 **Kuramoto model**: N coupled oscillators each with natural frequency ω_i; phase θ_i evolves as $\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_j \sin(\theta_j - \theta_i)$. Order parameter: $r = |\frac{1}{N} \sum_j e^{i\theta_j}|$, measuring synchrony ($r = 0$: incoherent; $r = 1$: full sync). Above a critical coupling K_c, a fraction of oscillators spontaneously synchronise. Example: firefly synchronisation, power-grid frequency alignment.

7.12 **Assortativity**: the tendency of nodes to connect to others that are similar to themselves. **Degree assortativity**: high-degree nodes preferentially connect to other high-degree nodes (positive $r > 0$) vs. high-degree nodes connecting to low-degree nodes (negative $r < 0$). Measured by Pearson correlation of endpoint degrees over all edges. **Positive (assortative)**: social networks — popular people know other popular people, forming a dense core-periphery structure. **Negative (disassortative)**: the Internet and biological networks — hubs connect to low-degree periphery nodes, producing a hub-and-spoke (star-like) structure.

7.13 A **giant component** is a connected component whose size is $O(N)$ (proportional to the entire network). **Network percolation**: in $G(N, p)$, as p increases from 0, small components form; at the **percolation threshold** $p_c = 1/(N-1) \approx 1/N$ (i.e., $\langle k \rangle \approx 1$), a giant component suddenly appears. Below this threshold: many small isolated components. Above: one dominant giant component coexists with small ones. This is a **phase transition** and relates to robustness: a network above threshold survives random node removal much better.

7.14 Diffusion model: node i has concentration c_i; substance flows along edges proportionally to concentration gradients.

$$\frac{dc_i}{dt} = \alpha \sum_{j \in N_i} (c_j - c_i)$$

Discrete update: $c_i(t + \Delta t) = c_i(t) + \alpha\Delta t\Big[\sum_{j\in N_i} c_j(t) - c_i(t) \cdot \deg(i)\Big]$. The network topology determines the diffusion rate: highly connected nodes equilibrate faster; isolated or peripheral nodes lag. Applications: species migration, currency flow, protein concentration in PPI networks.

Section 8 — ABM & Cellular Automata

8.1 CA = (S, s_0, N, d, f): S = finite state set; s_0 = initial configuration; N = neighborhood template; d = spatial dimension; $f : S^{|N|} \to S$ = local update rule applied synchronously. **Moore (2D)**: includes all 8 surrounding cells (cardinal + diagonal). **Von Neumann (2D)**: only 4 cardinal neighbors. Moore has more connections, sensitive to diagonal interactions; Von Neumann is simpler.

8.2 **Neither is universally preferable.** Moore neighborhood captures diagonal interactions and typically produces more isotropic spreading (e.g., realistic epidemic fronts). Von Neumann is simpler, more analyzable, and sufficient for problems with only cardinal directions. Choice depends on the physical/social reality being modeled. The statement is too absolute to be true in general.

8.3 $r = 1$: neighborhood size $= 2r + 1 = 3$; possible configurations $= 2^3 = 8$; number of rules $= 2^8 = 256$. $r = 2$: neighborhood size $= 5$; possible configs $= 2^5 = 32$; rules $= 2^{32} \approx 4.3 \times 10^9$.

8.4 Wolfram's classes:

- **Class 1**: all cells converge to a uniform state (fixed point).
- **Class 2**: periodic/simple structures (stable or oscillating patterns).
- **Class 3**: chaotic, aperiodic patterns filling space (no structure persists).
- **Class 4**: complex/localized structures, long transients, boundary between order and chaos — computationally universal (e.g., Rule 110).

8.5 **Game of Life** (Conway): 2D binary CA, Moore neighborhood. Rules: (1) live cell with 2 or 3 live neighbors survives; (2) live cell with < 2 dies (underpopulation); (3) live cell with > 3 dies (overpopulation); (4) dead cell with exactly 3 live neighbors becomes alive (reproduction). Static: "block" (2×2 square). Periodic: "blinker" (period-2). Glider: a pattern that moves diagonally across the grid.

8.6 **Schelling model**: agents on a grid, each with a type (e.g., red/blue). An agent is unhappy if fewer than a threshold t% of its neighbors share its type; unhappy agents move to a random empty cell. Result: even with mild preferences ($t \approx 30\%$), the system self-organises into strongly segregated clusters. This illustrates **emergence**: strong global segregation arises from weak individual preferences — a result that surprised social scientists when first shown by Schelling (1969).

8.7 ABMs capture: individual heterogeneity, stochastic decisions, spatial structure, emergent collective behavior. ODEs assume: homogeneous mixing, mean-field interactions, deterministic dynamics. ABM can reveal dynamics ODEs miss: phase transitions near critical thresholds, pattern formation, asymmetric outcomes from initially symmetric states, effects of network topology. Example: epidemic spreading on a clustered network vs. well-mixed SIR model — ABM shows slower onset and local clustering effects invisible to the ODE.

8.8 **ABM**: agents (prey/predator) move and interact locally; spatial patchiness, extinction corridors, and local heterogeneity are naturally captured. More appropriate when individual behavior or spatial effects matter. **Network**: models who interacts with whom but typically abstracts away spatial position; appropriate when the interaction topology (who knows whom) is the key feature, e.g., a food web with specific predation links.

8.9 Self-organisation: the spontaneous emergence of spatial or temporal order without a central controller, driven solely by local interactions. Example: **Rule 110** (Wolfram Class 4) produces persistent localised structures from a single non-zero cell — glider-like patterns emerge and interact, yet no cell has global knowledge. (Rule 30 also produces complex output from a single cell, but it is Class 3 — chaotic, not self-organised.) No agent "knows" the global pattern; order emerges from local rules applied simultaneously across all cells.

8.10 With $q = 3$ states and the rule depending only on the *two neighbors* (not the cell itself): the neighborhood is a pair (left, right), giving $q^2 = 3^2 = 9$ possible neighborhood configurations. Each configuration maps to one of $q = 3$ states. Number of rules $= q^{q^2} = 3^9 = 19683$.

8.11 Extract all neighborhood triples (left, center, right) $\to$ next center from the transition: Positions (1-indexed, with periodic or fixed boundaries): the key observable triples and their outputs are: $(0, 1, 0) \to 1$, $(1, 0, 0) \to 1$, $(0, 0, 1) \to 0$, $(0, 1, 0) \to 0$ — **contradiction!** Position 1: $(?, 0, 1) \to 0$. Position 3: $(0, 0, 1) \to 1$ vs. earlier $(0, 0, 1) \to 0$ — check carefully. Work out each position explicitly: cell i at $t + 1$ depends on (x_{i-1}, x_i, x_{i+1}) at t. Boundary cells use the boundary values shown. Map each triple to its output. If the same triple appears mapped to two different outputs, the transition is **inconsistent** (impossible for a deterministic rule). If consistent, the rule is defined by all encountered triples.

8.12 Majority rule: each node synchronously adopts the state held by the majority of its neighbors (if tie, stays or flips). **Absorbing states**: consensus (all 0 or all 1) as in the voter model, but also *frozen* configurations where no majority ever overturns the current state. **Voter model**: asynchronous, stochastic; a node copies one random neighbor. Both converge to consensus on finite connected networks, but: majority rule is deterministic and synchronous (may freeze in non-consensus states on some topologies); voter model is stochastic and always reaches consensus eventually.

8.13 Agent-based SIR: each agent has state $\in \{S, I, R\}$. Rules: (1) a susceptible agent who contacts an infected neighbor becomes infected with probability β; (2) an infected agent recovers (becomes R) with probability γ at each time step; (3) recovered agents are immune (no further transitions). Agents move in space (or interact on a network) and contacts happen locally. Advantages over ODE-SIR: (i) captures spatial heterogeneity (local outbreaks, geographic spread); (ii) models stochastic extinction — small populations can die out by chance, which ODEs miss; (iii) allows individual heterogeneity (age, mobility, compliance); (iv) can include multiple strains or co-morbidities naturally.

8.14 Extension for air travel: add **city nodes** (e.g., London, Lisbon) each containing a population of agents in the local ABM-SIR. A **travel network** layer (directed graph of flight routes) determines which cities are connected. At each time step a small fraction of agents at each city are probabilistically transferred to a destination city according to the flight network (sampling a destination by flight frequency/capacity). Newly arrived agents carry their current state (S, I, or R) into the destination's population and interact there. This enables: (i) **long-range seeding** — a local outbreak teleports to a distant city before local spread is detected; (ii) **network-structured global epidemics** — spread path follows the airline hub structure; (iii) studying intervention policies (travel bans modeled as removing edges from the flight network).

Section 9 — Graph Neural Networks

9.1 Each GNN layer performs: (1) **Message**: each node v sends a message $m_{uv} = M(h_u^{(k)})$ to neighbor v. (2) **Aggregate**: $a_v^{(k)} = \text{AGG}(\{m_{uv} : u \in N(v)\})$ (e.g., sum, mean, max). (3) **Update**: $h_v^{(k+1)} = U(h_v^{(k)}, a_v^{(k)})$ (e.g., MLP). After k layers, $h_v^{(k)}$ encodes information from the k-hop neighborhood of v.

9.2 **Transductive**: the model is trained and tested on the same fixed graph (new nodes require retraining). **Inductive**: the model learns a function that can be applied to unseen graphs or nodes. Transductive task: classifying nodes in a known citation network (GCN). Inductive task: predicting properties of new molecules not seen during training (GraphSAGE).

9.3 **GCN**: uses the full normalised adjacency matrix $\tilde{A}$ as the aggregation operator; computes $H^{(k+1)} = \sigma(\tilde{A}H^{(k)}W^{(k)})$. Requires the entire graph during training — *transductive*. **GraphSAGE**: samples a fixed-size neighborhood; learns a separate aggregator function (mean, LSTM, pooling). Can be applied to new nodes without retraining — *inductive*. Practical implication: GraphSAGE scales to large/dynamic graphs; GCN is simpler but inflexible.

9.4

- **Node classification**: predict a label for each node (e.g., classify research papers by topic in a citation network).
- **Link prediction**: predict whether an edge exists between two nodes (e.g., recommend friends in a social network, predict protein interactions).
- **Graph classification**: assign a label to an entire graph (e.g., classify molecules as toxic or non-toxic, classify brain connectivity patterns by disorder).